

SURIYAKALA S

Bioinformatics & Biotechnology Researcher

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Portfolio: suriyakala-s.github.io/Suriyakala-Portfolio/ | GitHub: github.com/Suriyakala-S | LinkedIn: linkedin.com/in/suriyakala07

PROFESSIONAL SUMMARY

Bioinformatics and Biotechnology professional with hands-on expertise in RNA-Seq, scRNA-Seq, variant calling, cancer genomics, and multi-omics data integration. Proficient in end-to-end NGS pipeline execution — from library preparation through differential expression and pathway enrichment — using R, Python, and Linux-based tools. Experienced in Oxford Nanopore Technologies (ONT) sequencing and wet-lab molecular biology techniques.

TECHNICAL SKILLS

Category	Tools & Technologies	Category	Tools & Technologies
Transcriptomics & Genomics	RNA-Seq, scRNA-Seq, Variant Calling, Cancer Genomics, DGE Analysis	NGS Tools	FastQC, HISAT2, STAR, SAMtools, Trimmomatic, Cytoscape, STRING
Sequencing Platforms	Oxford Nanopore (ONT), Illumina NGS, EPI2ME Workflows, Library Prep	Statistical Analysis	DESeq2, edgeR, limma, ggplot2, GO/KEGG/Reactome Enrichment
Programming & OS	R, Python, Linux/Unix, Bash Scripting	Multi-Omics	Metagenomics, AMR Profiling, Proteomics Integration, PPI Network Analysis
Molecular Biology	DNA/RNA Extraction, cDNA Synthesis, PCR, RT-PCR, Gel Electrophoresis	In Vivo Techniques	Drug Administration (Murine), Tissue Dissection, Cell Culture (H9c2, HepG2), ROS Assay

PROFESSIONAL EXPERIENCE

Junior Research Associate – Bioinformatics & Molecular Genetics | NuBiotech, Chennai *April 2026 – Present*

- Execute Nanopore long-read sequencing data analysis leveraging EPI2ME workflows for real-time basecalling, variant detection, and quality assessment.
- Lead transcriptomics and cancer genomics investigations, performing somatic variant calling and CNV analysis to identify clinically relevant genomic alterations.
- Conduct metagenomics profiling and antimicrobial resistance (AMR) gene characterization to support infectious disease and microbiome research initiatives.
- Design and execute end-to-end NGS and ONT sequencing library preparation protocols, ensuring optimal sequencing yield and data integrity.
- Perform molecular biology experiments including nucleic acid extraction, cDNA synthesis, PCR/RT-PCR amplification, and gel electrophoresis for downstream genomic applications.

Research Assistant – Bioinformatics & Scientific Writing | Vettri Biomed Technologies Pvt. Ltd. *October 2025 – March 2026*

- Analyzed bulk RNA-Seq and microarray expression datasets from GEO and TCGA repositories to identify molecular signatures in oncological and complex disease contexts.
- Implemented DGE analysis pipelines using DESeq2, edgeR, and limma with rigorous statistical filtering (adjusted p-value, log2 fold-change thresholds).
- Conducted GO, KEGG, and Reactome pathway enrichment analysis to elucidate dysregulated biological pathways and prioritize candidate therapeutic targets.
- Performed scRNA-Seq data interpretation including dimensionality reduction (UMAP/t-SNE), clustering, and cell-type annotation workflows.

- Engineered automated Linux-based bioinformatics pipelines, significantly reducing manual intervention and sequencing data processing turnaround time.

RESEARCH & PROJECT EXPERIENCE

Research Trainee – Cardiovascular & Metabolic Disease Genomics | THSTI, Faridabad December 2024 – April 2025

- Investigated mitochondrial gene expression profiles associated with CVD and NAFLD under the mentorship of Dr. Ruchi Tandon.
- Designed and executed in vivo experimental protocols in murine models including pharmacological compound administration, organ dissection, and biological sample collection.
- Performed RNA extraction from tissue biopsies and quantitative gene expression analysis to assess mitochondrial dysfunction and oxidative stress markers.
- Executed RT-qPCR assays, ROS quantification, lipid treatment studies, and functional experiments in H9c2 and HepG2 cell lines.
- Contributed to therapeutic compound screening workflows supporting pre-clinical evaluation of novel interventional strategies.

Research Intern – Nanomaterials & Antimicrobial Research | University of Madras May 2024 – June 2024

- Synthesized and characterized zinc oxide (ZnO) nanoparticles and evaluated antimicrobial efficacy in K-Carrageenan-based biofilm formulations against clinically relevant bacterial strains.

CERTIFICATIONS & TRAINING

NGS and Multi-Omics Data Analysis Internship | Biotecnika April 2025 – September 2025

- Completed training in bulk RNA-Seq, DNA-Seq, scRNA-Seq, and variant calling pipelines with quality control, alignment, and differential expression analysis.
- Integrated genomics, transcriptomics, and proteomics data layers for multi-omics studies identifying convergent molecular mechanisms.
- Capstone Project: Identified key regulatory genes associated with NAFLD through comprehensive RNA-Seq differential expression and pathway enrichment analysis.

EDUCATION

B. Tech in Biotechnology | Bharathidasan University, Tiruchirappalli 2021 – 2025
CGPA: 8.0 / 10.0

LANGUAGES

Tamil (Native) | English (Professional Working Proficiency) | Hindi (Basic Conversational)

REFERENCES

Dr. S. Sivaramakrishnan Department of Biotechnology Bharathidasan University, Tiruchirappalli +91 431 2407086 srkrishnan@bdu.ac.in	Dr. Sakthivel Muthu Assistant Professor Dept. of Dermatology Saveetha Medical College & Hospital, Chennai saktthivel@gmail.com	Dr. Premanand Adaikalasamy Managing Director & CEO Vettri Biomed Technologies Pvt. Ltd. +91 9363014568 director@vettribiomed.com
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